

rinna / **japanese-gpt-neox-3.6b-instruction-sft-v2**

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 Text Generation

 Transformers

 PyTorch

 Safetensors

 Anthropic/hh-rlhf

 stanfordnlp/SHP

 Japanese

 gpt_neox

 lm

 nlp

 text-generation-inference

 arxiv:2404.01657

 License: mit



 Train ▾

 Deploy ▾

 Use this model ▾

 **Model card**

 Files

 xet

 Community 1

Downloads last month

1,088



 **Safetensors** 

Model size 3.76B params

Tensor type F32 F16 U8

 Files info

Inference Providers NEW

 Text Generation

This model isn't deployed by any Inference Provider.

 Ask for provider support

Model tree for rinna/japanese-gpt-neox-3.6b-instruction-sft-v2

Base model rinna/japanese-gpt-neox-3.6b

 Finetuned (5) this model

 Quantizations 1 model

 Datasets used to train rinna/japanese-gpt-neox-3.6b-instruction-sft-v2

 Collection including rinna/japanese-gpt-neox-3.6b-instruction-sft-v2

japanese-gpt-neox-3.6b Collection

The japanese-gpt-neox-3.6b series are pr... • 5 items • Updated Aug 26 •  2

 **japanese-gpt-neox-3.6b-instruction-sft-v2**



[Overview](#)

This repository provides a Japanese GPT-NeoX model of 3.6 billion parameters. The model is based on [rinna/japanese-gpt-neox-3.6b](#) and has been finetuned to serve as an instruction-following conversational agent.

This model slightly differs from the previous SFT model [rinna/japanese-gpt-neox-3.6b-instruction-sft](#), where a different data split is used for training.

- **Model architecture**

A 36-layer, 2816-hidden-size transformer-based language model.

- **SFT vs. previous SFT evaluation**

We conducted ChatGPT-based automated evaluation on 100 prompts to assess the performance difference between this SFT model and the previous SFT model.

this SFT vs. previous SFT	win	tie	loss
ChatGPT auto. evaluation	55%	0%	45%

- **Finetuning**

The finetuning data is the subset of the following datasets and has been translated into Japanese.

- [Anthropic HH RLHF data](#)
- [FLAN Instruction Tuning data](#)
- [Stanford Human Preferences Dataset](#)

The data will **not** be released.

- **Model Series**

Variant	Link
3.6B PPO	https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-ppo
3.6B SFT-v2	https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-sft-v2
3.6B SFT	https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-sft
3.6B pretrained	https://huggingface.co/rinna/japanese-gpt-neox-3.6b

- **Contributors**

[Tianyu Zhao](#) and [Kei Sawada](#)

- **Release date**

March 31, 2023

I/O Format

A special format has been adopted to construct inputs.

- An input prompt is formatted as a conversation between ユーザー and システム.
- Each input utterance consists of (1) its speaker ("ユーザー" or "システム"), (2) a colon (":"), (3) a whitespace (" "), and (4) utterance text (e.g. "世界で一番高い山は？").

- The input prompt should be ended with "システム: " to acknowledge the model to generate a response.
- Since the model's tokenizer does not recognize "\n", a special newline symbol "<NL>" is used instead.
- All the newlines in input and output utterances should be replaced with "<NL>" .
- All the utterances in the input prompt should be separated by "<NL>" .

Following is an example to construct an input from a conversation.

```
prompt = [
    {
        "speaker": "ユーザー",
        "text": "コンタクトレンズを慣れるにはどうすればよいですか？"
    },
    {
        "speaker": "システム",
        "text": "これについて具体的に説明していただけますか？何が難しいのでしょうか？"
    },
    {
        "speaker": "ユーザー",
        "text": "目が痛いのです。"
    },
    {
        "speaker": "システム",
        "text": "分かりました、コンタクトレンズをつけると目がかゆくなるということですね。思った"
    },
    {
        "speaker": "ユーザー",
        "text": "いえ、レンズは外しませんが、目が赤くなるんです。"
    }
]
prompt = [
    f"{uttr['speaker']}: {uttr['text']}"
    for uttr in prompt
]
prompt = "<NL>".join(prompt)
prompt = (
```

```

prompt
+ "<NL>"
+ "システム: "
)
print(prompt)
# "ユーザー: コンタクトレンズを慣れるにはどうすればよいですか？<NL>システム: これについて具体的に

```

🔗 How to use the model

```

import torch
from transformers import AutoTokenizer, AutoModelForCausalLM

tokenizer = AutoTokenizer.from_pretrained("rinna/japanese-gpt-neox-3.6b-instructic
model = AutoModelForCausalLM.from_pretrained("rinna/japanese-gpt-neox-3.6b-instruc

if torch.cuda.is_available():
    model = model.to("cuda")

token_ids = tokenizer.encode(prompt, add_special_tokens=False, return_tensors="pt"

with torch.no_grad():
    output_ids = model.generate(
        token_ids.to(model.device),
        do_sample=True,
        max_new_tokens=128,
        temperature=0.7,
        repetition_penalty=1.1,
        pad_token_id=tokenizer.pad_token_id,
        bos_token_id=tokenizer.bos_token_id,
        eos_token_id=tokenizer.eos_token_id
    )

output = tokenizer.decode(output_ids.tolist()[0][token_ids.size(1):])
output = output.replace("<NL>", "\n")
print(output)
"""わかりました。まずは、コンタクトレンズを長時間着用することによる目の乾燥を防ぐことができます。

```

🔗 Tokenization

The model uses a sentencepiece-based tokenizer.

- The tokenizer has a vocabulary size of 32,000.
- It uses sentencepiece's byte fallback feature to decompose unknown text pieces into UTF-8 byte pieces and to avoid producing `<UNK>` tokens.
- sentencepiece's `--add_dummy_prefix` option was turned off so that a leading whitespace will not be prepended automatically.

```
print(tokenizer.tokenize("吾輩は猫である"))  
# ['吾', '輩', 'は', '猫', 'である']  
# instead of ['_', '吾', '輩', 'は', '猫', 'である'] as in rinna/japanese-gpt-
```

- sentencepiece's `--remove_extra_whitespaces` option was turned off so that leading, trailing, and duplicate whitespaces are reserved.

```
print(tokenizer.tokenize(" 吾輩は 猫である  "))  
# ['_', '_', '吾', '輩', 'は', '_', '_', '猫', 'である', '_', '_', '_']  
# instead of ['_', '吾', '輩', 'は', '_', '猫', 'である'] as in rinna/japanese-gpt-
```

- Don't forget to set `use_fast=False` to make the above features function correctly.

```
good_tokenizer = AutoTokenizer.from_pretrained("rinna/japanese-gpt-neox-3.6b")  
bad_tokenizer = AutoTokenizer.from_pretrained("rinna/japanese-gpt-neox-3.6b")  
  
print(good_tokenizer.decode(good_tokenizer.encode("გამარჯობა 吾輩は 猫である"))  
# 'გამარჯობა 吾輩は 猫である </s>'  
print(bad_tokenizer.decode(bad_tokenizer.encode("გამარჯობა 吾輩は 猫である"))  
# 'გამარ[UNK]ობა 吾輩は 猫である </s>'
```

🔗 How to cite

```
@misc{rinna-japanese-gpt-neox-3.6b-instruction-sft-v2,  
  title = {rinna/japanese-gpt-neox-3.6b-instruction-sft-v2},  
  author = {Zhao, Tianyu and Sawada, Kei},  
  url = {https://huggingface.co/rinna/japanese-gpt-neox-3.6b-instruction-sft-v2}  
}  
  
@inproceedings{sawada2024release,  
  title = {Release of Pre-Trained Models for the {J}apanese Language},  
  author = {Sawada, Kei and Zhao, Tianyu and Shing, Makoto and Mitsui, Kentaro a  
  booktitle = {Proceedings of the 2024 Joint International Conference on Computa  
  month = {5},  
  year = {2024},  
  pages = {13898--13905},  
  url = {https://aclanthology.org/2024.lrec-main.1213},  
  note = {\url{https://arxiv.org/abs/2404.01657}}
```

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